3D-Attentioned-UNet for Alzheimer disease diagnosis

Detect and classify salient features across three different brain conditions.

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Abstract

Alzheimer's disease is very insidious and difficult to recognize. From a general medical point of view, dementia can occur in many different forms, which are not always due to Alzheimer's disease. Individuals who develop a state of mild cognitive impairment have characteristics that bring them closer to CN patients or AD patients. This can be inferred from the fact that Alzheimer's disease in its most advanced stage is diagnosed many decades later than the onset of brain damage, its incubation period is very long and its significant local features are not always accurately recognized. Therefore, the age of the patient plays a crucial role in the diagnosis. In this experiment, I will try to diagnose the disease by training a model that can distinguish data from three different categories: CN, MCI and AD. The logic of the model is based solely on the characteristics of the brain, the age and the gender of the patient. In this way, I deviate as far as possible from the approach of diagnosing the disease based on biological indicators and neuropsychological tests evaluated by a human being.

Keywords

PET: Positron Emission Tomography, **MRI**: Magnetic Resonance Imaging, **AD**: Alzheimer Disease, **MCI**: Mild Cognitive Impairment, **CN**: Cognitive Normal, **DSP**: Deep Supervision, **FSL**: FMRIB Software Library, **BET**: Brain Extraction Tool, **MMSE**: Mini-Mental State Examination, **EMCI**: Early Mild Cognitive Impairment, **LMCI**: Late Mild Cognitive Impairment,

HIGHLIGHTS

- 1: Propose a different approach in terms of classification particularly influenced by the nature of the data: instead of a binary classification between couple of brain states [1],[2], the model set the base for a classification between three different classes.
- 2: Exploiting the advance of using age and sex as addition information to characterize the brain features during the classification process.
- 3: breaking down the model's structure to add attention mechanism in order to level out most relevant features instead of low relevance details.
- 4: Make use only of "biological elements" such as brain features, age and gender without taking into account any other human contribution such as MMSE.

1. Introduction

Dementia from the Latin de mens, *memory gone*, refers to a class of diseases that impair the patient's mind. There are several forms of dementia that have a greater or lesser impact on the patient's life: Alzheimer's disease is one of the most common forms of senile dementia. To date, it affects 5% of people over the age of 60, and in Italy there are an estimated 500,000 patients diagnosed with Alzheimer's disease. [3, 4]

The first documented case of Alzheimer's disease reported by Alois Alzheimer was that of Mrs. Auguste Deter, a 51-year-old woman who in 1901 showed signs of disorientation, delusions and cognitive disturbances that led to her death after 4 to 5 years. Alzheimer's neuropathological examination revealed diffuse brain atrophy and peculiar changes in cortical cell clusters. Alzheimer was able to provide a qualitative overview, but it was not until the 1980s that the disease was hypothesized to be caused by the presence of plaque $\beta - amyloid$ and hyperphosphorylated τ protein in neurofibrillary tangles. [5]

Alzheimer's is a **progressive disease**, which means that more neurons and more areas of the brain are damaged over time. The disease begins with damage to neurons in areas responsible for language (`Temporal` cortex), memory storage and thought processes (`Pre-Frontal` and `Frontal` cortex with extension to the `Hippocampus`). In addition to progressive memory impairment, patients may also develop changes in mood, behavior or personality [3]. The symptoms and the speed of their progression can vary from person to person: patients over the age of 65 who develop the disease survive an average of four to eight years, as the neuronal damage can also affect areas of the brain responsible for maintaining basic bodily functions, eventually leading to the patient's death. [6]

Alzheimer's disease diagnosis is difficult for several reasons:

- The symptoms described are common to many

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types of dementia and may overlap with comorbidities, making the disease difficult to diagnose, especially in the early stages.

- The pathophysiology of Alzheimer's dementia (Alzheimer's-related brain changes) begins years or sometimes **decades** before diagnosis. Local changes are sometimes imperceptible and the disease is revealed only when in advanced stages.
- The diagnosis is based on the clinical history and the patient's presentation (current status). In fact, the exclusive use of imaging, biomarkers and PET tests is not sufficient to rule out or rule in Alzheimer's disease in the majority of symptomatic individuals. [7]

PET scans can provide information about the amyloid burden over the course of a patient's life. In particular, data from clinically normal elderly individuals suggest that the presence of biomarkers of $\beta - amyloid$ accumulation is associated with functional and structural changes in the brain that are consistent with abnormalities observed in patients with MCI and Alzheimer's disease. However, many histopathologic features can only be observed postmortem. [8]. The latter is due to the fact that many affected patients never show behavioral symptoms during their lifetime; in these cases, the diagnosis is made at autopsy. [9]

AD Progression Alzheimer's disease progresses through three different stages:

The first is the **preclinical phase**. Patients in this phase have no symptoms at first. The phase begins with asymptomatic cerebral amyloidosis. Cognitive deficits do not yet occur, but in the last stage of the pre-clinical phase there is a measurable reduction in the patient's cognitive performance.

The second stage of the disease manifests in a **mild cognitive decline**. During this phase, despite the patient shows signs of cognitive decline, is still capable to be autonomous. The MCI due to AD is diagnosed through cognitive and functional tests. In any case, the negativity of biomarkers at this stage suggests that AD pathology is not the cause of the patient's deterioration.

Alzheimer's disease dementia is the third and final stage. It is characterized by a gradual onset of cognitive or behavioral symptoms that begin to impair the patient's daily life. The diagnosis of dementia must include impairment in at least one of the following domains: memory storage, reasoning, visuo-spatial abilities, language, or changes in personality or behavior. [8]

Biomarkers are measurable indicators in a patient that can help to assess and mark the progression of Alzheimer's disease. They can be divided into two main classes: direct and indirect. The direct ones, such as $\beta - amyloid$ protein and τ, are specific key proteins that reflect the pathology. The τ can be measured in cerebrospinal fluid (CSF) and is a strong marker of neuronal and synaptic damage and is specifically associated with Alzheimer's disease. PET scans may also show decreased uptake of $18 - fluorodeoxyglucose$, while structural MRI may show disproportionate atrophy. In particular, high-resolution MRI is able to differentiate the smallest structures of the brain, while PET has enabled staging of ante-mortem Alzheimer's disease.

Neuropsychological test The Mini-Mental State Examination (MMSE) is a widely used neuropsychological test to assess cognitive impairment and is frequently used in the diagnosis and monitoring of Alzheimer's disease and other forms of dementia. Developed by Folstein et al. in 1975, the MMSE is a brief screening instrument that assesses several cognitive domains, including orientation, memory, attention, language and visuospatial abilities. It consists of a series of questions and tasks. The total score ranges from 0 to 30 points, with lower scores indicating greater cognitive impairment. A score of 18 or less indicates severe cognitive impairment; a score between 18 and 24 indicates moderate to mild impairment, a score of 25 is considered borderline, and 26 to 30 indicates cognitive normality.

Although the MMSE is useful for quickly assessing cognitive function, it is not in itself a definitive diagnostic tool. A comprehensive assessment usually involves a combination of clinical interviews, a review of medical history, neuroimaging and other neuropsychological tests.

2. Neural model for AD Diagnosis

Building a neural model for the diagnosis of Alzheimer's disease can prove to be a difficult task, mainly due to the anatomapathology of the disease and the type of input data I will use to train our model.

To diagnose Alzheimer's disease, the model needs to interact with 3D brain images (MRIs) that have two important parameters associated with them: the patient's gender and age. This leads us to an important first assumption for our task: since it's dealing with human's brains, they will be similar in different people of the same age and gender.

A second pitfall lies in the nature of classification: the model is called upon to classify human brains into three categories: CN, MCI, AD.

The literature shows that most people diagnosed with MCI tend to develop AD above a certain age limit, while others do not. This leads us to another assumption: given the morphological nature of the human brain, brains classified as MCI that are "mistaken" for CN and AD inher-

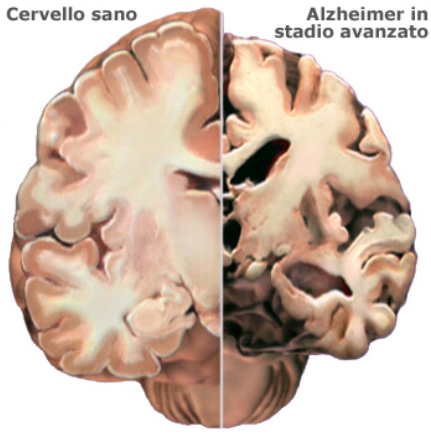


Figure 1: A healthy brain compared to one suffering from Alzheimer's disease. The crumpled brain tissue enlarges the space between the brain ventricles and enlarges the empty brain regions, which can be seen as suspicious black regions on the MRI image of the brain. (https://www.alz.org/brain_italian/09.asp).

ently have local features similar to those of individuals with Alzheimer's disease or normal individuals.

This assumption is important because, on the one hand, it lays the foundation for a possible working methodology for **"diagnosis of possible disease progression"**, but on the other hand, it represents an important constraint for our convolutional architecture for feature learning, as it could be confusing due to the presence of local features that are a specificity of individuals with or without Alzheimer's disease.

In practice, Alzheimer's disease in its most advanced stage manifests itself in a crumpling of the brain tissue. As can be seen in Figure.1, the empty space between the ventricular cavities of the brain increases due to the thinning of the brain tissue.

3. Data

3.1. Dataset

The datasets provided by ADNI were selected for the experiment. The collected data are divided into 4 different sets: One set refers only to the screening phase: it includes the visits of patients who participated only and exclusively in the screening phase, without further medical monitoring afterwards. Three datasets (first year, second year and third year) include the visits of patients who participated in the screening phase and continued monitoring. A very important indication is that for most patients there is no significant change in the morphological state of the brain over the 3-year period; this means that the diagnosis of the disorder does not evolve at the level of the individual patient.

3.2. Data Experiment

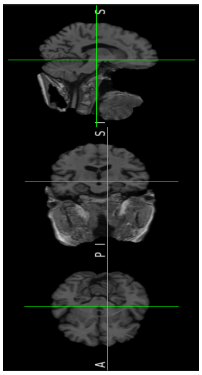
In order to conduct the experiment more smoothly, I decided to split the data as follows: subjects who only participated in the screening phase will be isolated in a separate dataset that will serve as `test.db`. This is to ensure that we have a dataset that is completely separate from the monitoring logic used to generate the other data along the 3-year ADNI timeline. The data from patients who have continued monitoring are instead combined into another dataset, split into `train.db` and `valid.db`, which is used for training the model. In this scenario, we need to make some considerations: The dataset contains several samples that can be considered redundant since the patients are humans and most of the brains belonging to the same class are quite similar accounting age and sex. Moreover, the disease does not evolve at the level of the individual patient during the three years of monitoring.

4. Data Processing

As reported in the studies by Fan et al. and Qiu et al., to diagnose the disease using MRI images, these images must be cleaned of details that are irrelevant to the diagnosis: the most important element is the brain tissue, while everything else, especially the skull, must be eliminated. For this purpose, I used the tool BET from FSL[1][1], or alternatively is possible to rely on FreeSurfer[2]. Through some experiments, I found that FreeSurfer from Hoopes et al. is much more accurate than FSL in eliminating everything that is not brain tissue from the MRI images. This is due to the way their neural network is trained: as the authors report, the images go through a **augmentation pipeline**. During this process, new samples are generated from a single segmentation map, spatial variability is induced by applying random nonlinear transformations, new voxel values are sampled from a Gaussian distribution, blurring, downsampling and various corruptions are applied to make a single sample more diverse. This increases the generalization capability of the network compared to previous approaches such as the simple FSL library.
The data processing stage is crucial to the work: since the disease itself is already difficult to diagnose, the recurrent presence of redundant features may cause the neural network to focus on extracting a different type of feature that has nothing to do with the disease. This would leads to a general difficulty in training the model, resulting in a rapid deterioration of performance.

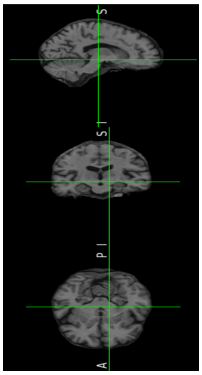
[1] Oxford Centre for Functional MRI of the Brain, https://fsl.fmrib.ox.ac.uk/fsl/fslwiki/FSL

[2] https://surfer.nmr.mgh.harvard.edu/



(a) FSL MRI Processing

(b) FSL resulting brain 3D reconstruction



(c) FreeSurfer MRI Processing

(d) FreeSurfer resulting brain 3D reconstruction

Figure 2: MRI reading and 3D Brain Reconstruction with FSLView: it can be seen from (a) and (b) that using the FSL library alone to perform skull stripping leads to unsatisfactory results in terms of removing anything that is not brain tissue. Using the FreeSurfer library with the `SynthStrip` model [10] leaves only the patient's brain uncovered (c) and (d), thus removing all parts from the MRI that might tempt the neural network to focus on other local features.

5. Related Works

Previous researches The architecture chosen to solve the task relies on U-Net [11], a convolutional neural model that is very useful for processing medical images. U-Net by Ronneberger et al., is an architecture consisting of two main elements, an Encoding block and a Decoding block, separated by a sequence convolutional blocks at the bottom of the structure (this gives it its U-shape). In the encoding block, the image flows through a series of convolutional blocks separated by a Batch Normalization operation and a DownSampling operation, which progressively reduces the size of the feature maps extracted at the upper convolutional levels. In the decoding block, features from the lower levels, which are linked to features from some of the encoder levels (skip connections) by concatenation, pass through a series of convolutional blocks, which are also separated from each other by a Batch Normalization and an UpSampling operation. This gradually increases the size of the feature maps until a segmented output is returned, that is ultimately used for 3D image classification.

Fan et al. in another research observed that basic u-net model is not comfortable with classification task, since it is designed to do segmentation (output an image of the same size with labels). They proposed an interesting review of the original model to the classification task [2].

6. Architecture

New Approach The architecture that was built to solve the task is strongly influenced by the research of Fan et al.. The model proposed in Figure.3 adds two very important elements: two embedding representations for the patient's gender and age, which serve as additional information to the brain features extracted from the MRI by convolutions, and three attention blocks at the level of skip connections, a method used in two other studies by Oktay et al. and Agarwala et al. that use convolutional networks as well to detect pancreatic and brain tumors.

6.1. Attention during Convolutional Process

For our Alzheimer's classification process, it is crucial to introduce an element that induces some kind of memory mechanism: this will help the model learn to omit feature sequences that may not be very frequent in favor of feature sequences that are more frequent in the samples of our training set. In general, RNN or Transformer type architectures can be used to model a memory mechanism; Since the input data and the resulting convolutional features are very memory intensive, in this particular case I did not rely on U-Net+RNN or U-Net+Transformer type architectures to learn the most recurrent features across images, I decided to refer to two other research papers, dealing with brain tumor detection [13] and pancreatic tumor detection [12], which apply **an attention mechanism** at the level of skip conncections to allow the model to better orient between recurrent and non-recurrent features.

As shown in Figure.4, attention is designed to augment the receptive field of the model and thus capture more semantic contextual information. The attention coefficients capture salient regions from the brain features and attempt to retain only the activations relevant to the task. g represents the "gating signal vector" from the lower convolutional layer, which contains lower level feature pruning information, while x^l represents the feature vector for the skip connection. g is used to determine the

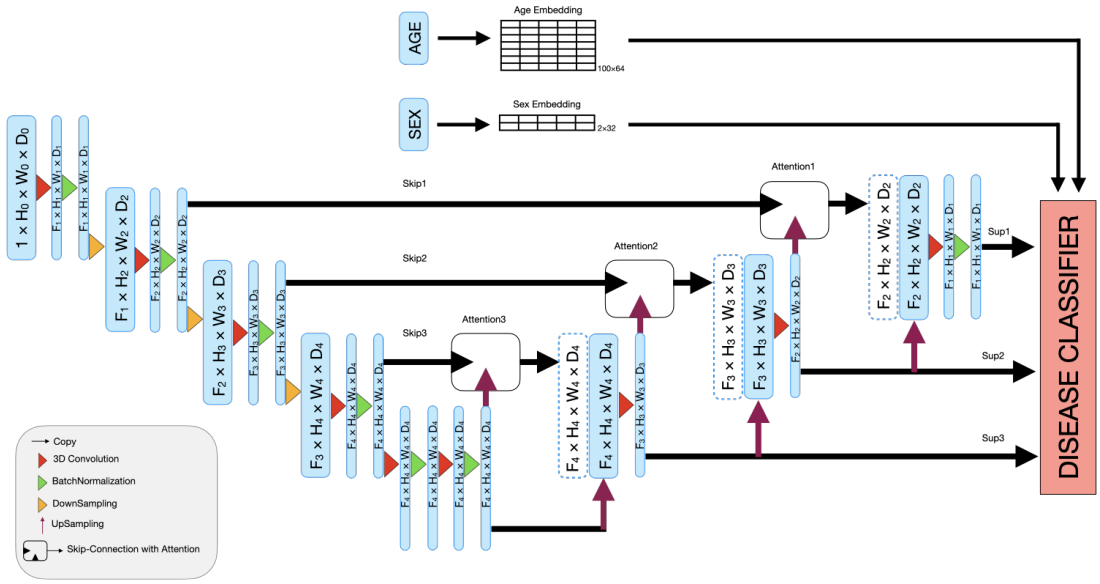


Figure 3: Architecture: A revised U-Net model with Deep Supervision Classification and attention blocks at the skip connection level. In addition, two embeddings are introduced, one for gender and one for age, to better characterize the brain features that enter the disease classifier

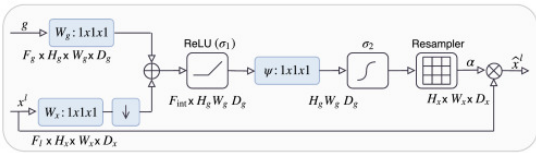


Figure 4: 3D Attention. Attention coefficients α are used to focal the model attention on the most relevant region of the feature map x^l (https://ars.els-cdn.com/content/image/1-s2.0-S1361841518306133-gr2.jpg).

focal region in the feature vector x^l and both give their contribution in creating the attention coefficients:

$$\alpha = \sigma_2(\psi(\sigma_1(W_g g + W_x x^l))) \tag{1}$$

Attention coefficients are thus used to focal the model on the most salient region of the feature map x^l:

$$\hat{x}^l = \alpha \times x^l \tag{2}$$

6.2. Increase information rate with age and sex

The patient's age and gender are coded using two embedding matrices:

- Age Embedding: is a 100×64 matrix that encodes the patient's age in a latent vector of dimension 64. The dimension 100 is chosen to simulate a `age line` from 1 to 99 years. The basic assumption is that since the dataset consists of patients whose age varies approximately between 55 and 94, the emebeddings corresponding to these ages will be most influenced by training to capture similarities between the different age characteristics with respect to the patient's disorder classification provided by the model.
- Sex Embedding: similar to `age embedding`, this is a 2×32 matrix that encodes the patient's sex in a latent vector of dimension 32.

The latent dimensions of the two matrices are empirically selected and calibrated to have a higher dimension for age, as the range of values is significantly higher than for gender, which is binary, i.e. M or F. The main function of these embeddings is to add additional information to the feature vector that is entered into the classifier: the 1D vectors are expanded in two additional directions to form a matrix and concatenated with the feature vector. In this way, gender and age labels are added to the features in some way to characterize them more precisely. This has the effect of initially attenuating the similarity of the patients' brains, separating the male subjects from the female subjects and adding the basic component of age, which is a clearer distinguishing detail for the possible occurrence of the disorder.

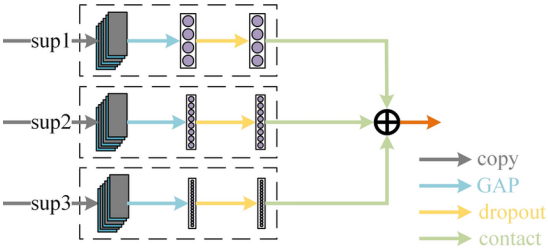


Figure 5: Deep Supervision block: its function is essentially to combine features from the model's skip connections to complete the classification, taking into account local features that can be found in subjects at deeper convolutional levels (**high level features**). (https://www.researchgate.net/figure/DSP-block-which-shows-the-detail-of-deep-supervision_fig3_351036583).

6.3. Disease Classifier

Regarding the disease classifier, I have chosen a similar approach to the one proposed by the authors Fan et al. in their research, which benefits from the `attentioned skip connections` in combination with the additional information contained in the sex and age embeddings: the DSP (Deep Supervision)[2] block is shown in Figure.5: after concatenating the features of the 3 skip connections, the patient's gender and age, which were coded via the embedding matrices, are also added. This step, as already mentioned, uniquely characterizes the block of features that is entered into the classifier. This provides the network with additional knowledge (not only about the patient's brain features) and significantly stabilizes the training process. In this way, the observable results from the learning curves show how it increases the overall classification capability of the model in terms of accuracy.

7. Experiment

This section illustrates the key design decisions made to accomplish the task. The experiment consists of three different versions of the model:

- Base model: is simply the implementation of u-net, adapted to our domain of Alzheimer's detection.
- Advance model: increases the capability of the model by adding information to the brain features using age and gender embeddings.
- More Advance model: exploit the capability of the attention mechanism applied to the advance model, resulting in the final **3D-Attentioned-UNet**.

7.1. Design Choices

In contrast to the existing literature, in which authors usually try to distinguish between CN and AD, in this experiment I have trained a model that distinguishes between CN, MCI and AD. The task is inherently more complex due to the similarity considerations of the input data: in particular, subjects classified as MCI have a percentage imbalance in local features that bring them closer to a CN or AD classification. This makes the network more susceptible to the local features themselves and leads to confusion, which is reflected in fluctuations in classification rates between the three labels. Looking closely at the confusion matrix in Figure.??, it can be seen that the pattern is more confused for the pairs (MCI,CN) or (MCI,AD) than for (CN,AD). The MCI patterns also complicate the diagnosis of the AD class, as the disorder in its most advanced stage may take different lengths of time to become more manifest, depending on the age of the patient and the degree of local features characterizing the disease.

7.2. Training strategy

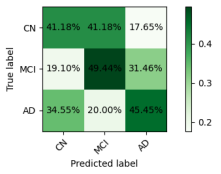
Since I had to use 3D convolutions in order not to lose the spatiality of the features, and since I was dealing with particularly huge images, I had to use some tricks to set up the training of the model correctly: due to hardware limitations, I chose to use a unitary batch size; this allows the model to be trained, but does not allow direct comparison between different training samples within the time windows of each epoch. All this has a strong impact on the generalization ability of the model. To overcome this limitation, the gradient accumulation strategy was chosen to be able to simulate a larger batch size. Although both Qiu et al. and Fan et al. do not really seem to aim for a higher order batch size in their research, I observed how it's important to let the model to propagate the gradient over a larger number samples to constast the redundancy of the data due the condition of the experiment, augment the generalization capability and better train the embedding matrices referred to age and gender.

8. Results

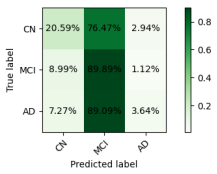
As can be seen from Table.1, the model that performs best on an unseen dataset is in fact the `more advanced` model that exploits both the sex and age mechanism and the attention at skip connections level. The `base model` achieves the worst performance. From the curves in Figure.?? it's possible to see as the accuracy on the training set increases uncontrollably, while the accuracy on the validation set continues to decrease. The `advance model` represents an intermediate path and shows how

Table 1
Training Results: base model refers to the primary re-implementation of u-net architecture without other improvements; Advance model exploit the potentiality of using sex and age embeddings to better characterize the brain features during the classification; More Advance model indeed, put all the element together in a new architecture, 3D-Attentioned-UNet

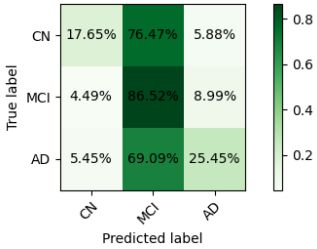
Architecture	Train F1	Valid F1	Test F1
`Base model`	0.91	0.55	0.47
`Advance model`	0.79	0.55	0.50
`More Advance model`	0.59	0.52	0.55



(a) Base model test



(b) Advance model test



(c) More Advance model test

Figure 6: Result of the test for `test.db`, which is composed starting by the patients who only complete the screening phase without continuing with monitoring.

the use of age and sex embeddings stabilizes the training curves, which start to follow a similar trend.

9. Hints of Explainability

In general, with the tools available, it is difficult to provide a transparent interpretation of the processes that take place within a neural network and cause the model to produce a particular output. Since I'm dealing with a task as insidious and complex as the diagnosis of Alzheimer's disease, I believe it is our duty to provide the reader with a clearer overview of the steps the neural network takes to classify a particular patient: the model essentially creates a probability distribution of brain features to recognize similar patterns among different individuals exhibiting a particular mental state. This probability distribution leads the model to characterize areas that are common to all patients, areas of intermediate specificity that are more manifest in patients in the Mild Cognitive Impairment (MCI) class, to areas of greater specificity that represent the most common features of patients in the advanced Alzheimer's Disease (AD) class.

9.1. Gradient CAM

To get a better insight into the classification process of the model, I used a technique known as "Gradient CAM": Gradient-weighted Class Activation Mapping (Grad-CAM) is a visualization technique used to illustrate the predictions of deep convolutional neural networks (CNNs). It works by generating a heatmap that highlights the regions of an input image that are most relevant to a given prediction. Grad-CAM is based on the observation that the gradients of the loss function with respect to the features of the final convolutional layer can be used to localize the discriminative regions of the input image.

$$CAM_c = ReLU\left(\sum_k w_c^k A_k\right) \tag{3}$$

where:

- `ReLU(x)` is the rectified linear unit activation function, defined as $\max(0, x)$.
- w_c^k is the weight of the k-th feature map for class c.
- A_k is the k-th feature map.

After the computation, CAM need to be normalized in the range $[0, 1]$:

$$CAM_c' = \frac{CAM_c}{\max(CAM_c)} \tag{4}$$

10. Conclusions

The results of the experiment with this new type of model are overall lower than in the existing bibliography, but taking into account that the model classifies on three output labels of the disease and not in a binary way. It would be better to increase the number of output labels by capping the samples in the MCI class and dividing them into EMCI and LMCI to increase the model's classification power. However, the learning curves still give reason to hope that the model can improve its performance if it is trained more. Overall, the research shows a renewed approach from the experiment with the attention mechanism integrated into the model to recognize the most important features for the diagnosis of Alzheimer's disease.

References

[1] S. Qiu, P. S. Joshi, M. I. Miller, C. Xue, X. Zhou, C. Karjadi, G. H. Chang, A. S. Joshi, B. Dwyer, S. Zhu, M. Kaku, Y. Zhou, Y. J. Alderazi, A. Swaminathan, S. Kedar, M.-H. Saint-Hilaire, S. H. Auerbach, J. Yuan, E. A. Sartor, R. Au, V. B. Kolachalama, Development and validation of an interpretable deep learning framework for Alzheimer's disease classification, Brain 143 (2020). doi:10.1093/brain/awaa137.

[2] Z. Fan, J. Li, L. Zhang, G. Zhu, P. Li, X. Lu, P. Shen, S. Shah, M. Bennamoun, T. Hua, W. Wei, U-net based analysis of mri for alzheimer's disease diagnosis, Neural Computing and Applications 33 (2021) 13587–13599. doi:10.1007/s00521-021-05983-y.

[3] I. S. di Sanità, Malattia di alzheimer, ???? URL: https://www.epicentro.iss.it/alzheimer/.

[4] C. Turkington, D. Mitchell, The Encyclopedia of Alzheimer's Disease, Facts on File Library of Health & Living, Facts On File, Incorporated, 2010. URL: https://books.google.it/books?id=SA2X3ZHUZaEC.

[5] J. A. Soria Lopez, H. M. lez, G. C. ger, Alzheimer's disease, Handb Clin Neurol 167 (2019) 231–255.

[6] 2022 alzheimer's disease facts and figures, Alzheimer's & Dementia 18 (2022) 700–789. URL: https://alz-journals.onlinelibrary.wiley.com/doi/abs/10.1002/alz.12638. doi:https://doi.org/10.1002/alz.12638.

[7] A. Atri, The Alzheimer's Disease Clinical Spectrum: Diagnosis and Management, Med Clin North Am 103 (2019) 263–293.

[8] F. T. Hane, M. Robinson, B. Y. Lee, O. Bai, Z. Leonenko, M. S. Albert, Recent Progress in Alzheimer's Disease Research, Part 3: Diagnosis and Treatment, J Alzheimers Dis 57 (2017) 645–665.

[9] R. A. Sperling, P. S. Aisen, L. A. Beckett, D. A. Bennett, S. Craft, A. M. Fagan, T. Iwatsubo, C. R. Jack, J. Kaye, T. J. Montine, D. C. Park, E. M. Reiman, C. C. Rowe, E. Siemers, Y. Stern, K. Yaffe, M. C. Carrillo, B. Thies, M. Morrison-Bogorad, M. V. Wagster, C. H. Phelps, Toward defining the preclinical stages of Alzheimer's disease: recommendations from the National Institute on Aging-Alzheimer's Association workgroups on diagnostic guidelines for Alzheimer's disease, Alzheimers Dement 7 (2011) 280–292.

[10] A. Hoopes, J. S. Mora, A. V. Dalca, B. Fischl, M. Hoffmann, Synthstrip: skull-stripping for any brain image, NeuroImage 260 (2022) 119474. URL: https://www.sciencedirect.com/science/article/pii/S1053811922005900. doi:https://doi.org/10.1016/j.neuroimage.2022.119474.

[11] O. Ronneberger, P. Fischer, T. Brox, U-net: Convolutional networks for biomedical image segmentation, 2015. URL: http://arxiv.org/abs/1505.04597.

[12] O. Oktay, J. Schlemper, L. L. Folgoc, M. Lee, M. Heinrich, K. Misawa, K. Mori, S. McDonagh, N. Y. Hammerla, B. Kainz, B. Glocker, D. Rueckert, Attention u-net: Learning where to look for the pancreas, in: Medical Imaging with Deep Learning, 2018. URL: https://openreview.net/forum?id=Skft7cijM.

[13] S. Agarwala, S. Sharma, B. U. Shankar, A-unet: Attention 3d unet architecture for multiclass segmentation of brain tumor, 2022. doi:10.1109/TENSYMP54529.2022.9864546.

A. Confusion Matrices

The confusion matrices for the most advanced model are shown here. It can be see from Figure.?? that the model is still confused with respect to the prediction of the MCI class, which is mainly predicted as CN and AD. On the one hand, this represents a gap that the model needs to fill in terms of performance with more training, but summarizing the reasoning so far, the similarities of the features of the MCI class to those that would theoretically bring them closer to the CN or AD classes also play an important role by showing that they might actually be individuals suffering from some form of dementia but not fully attributable to Alzheimer's disease.

B. Gradient CAM Results

Figure.11 shows what the model captured as the regions of greatest interest for the diagnosis of possible Alzheimer's disease. It can be noted that, consistent with what has been said, the areas of greatest intensity are those affecting the hippocampus, with an extension to the frontal and temporal areas.

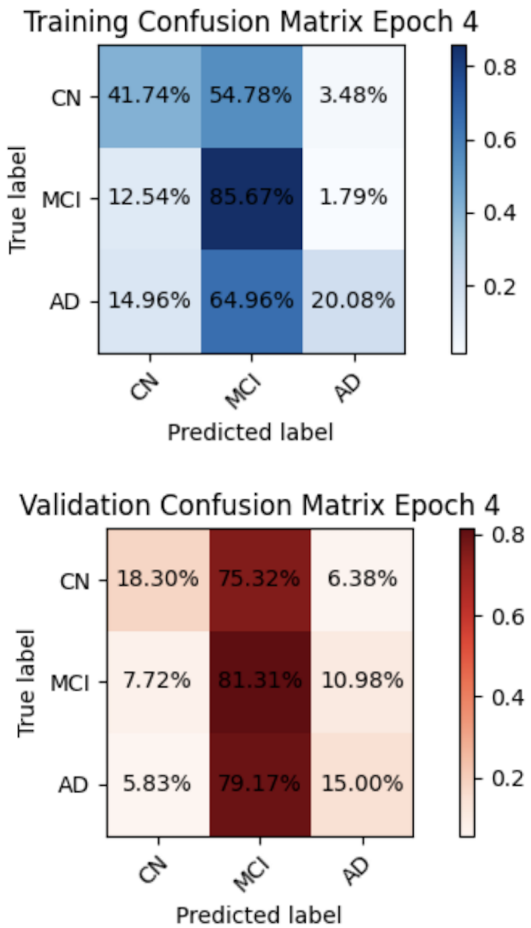


Figure 7: Training and Validation Confusion Matrix of 3D-Attentioned-UNet.

Figure 8: Male Subject of 82 years old which suffer of AD

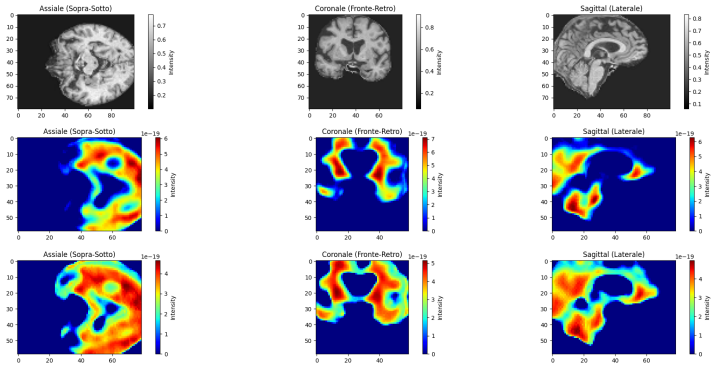


Figure 9: Male Subject of 74 years old, CN

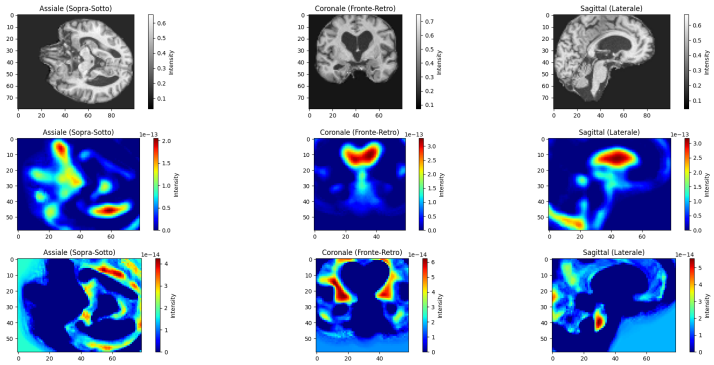


Figure 10: Male Subject of 58 years old which suffer of MCI

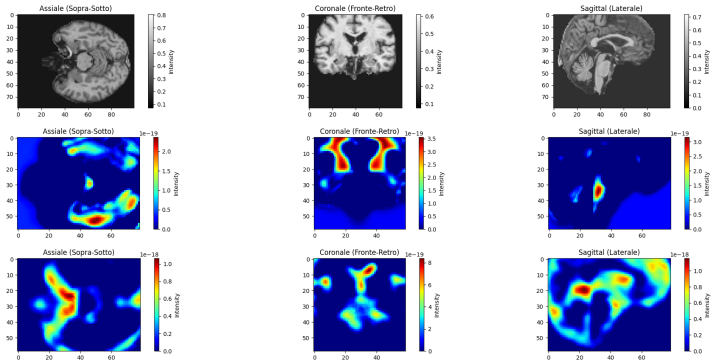


Figure 11: Male Subject of 79 years old which suffer of MCI

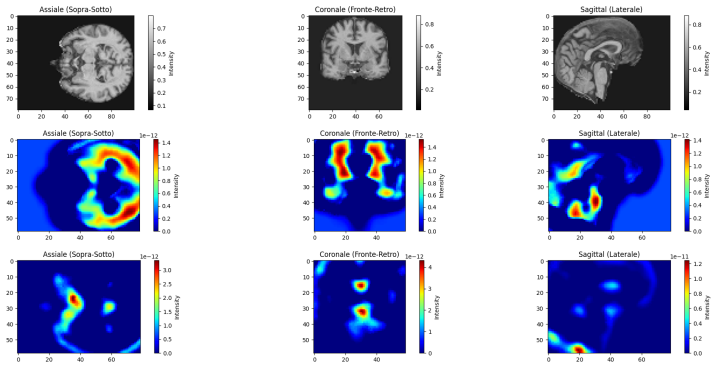
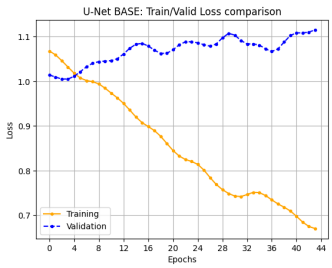
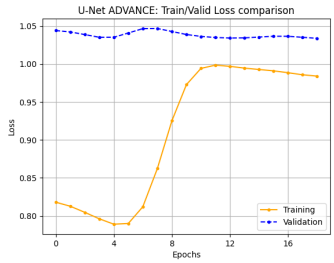


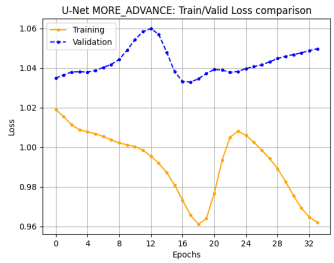
Figure 12: Gradient Cam: most salient region that contribute the model prediction of the subject's mental state



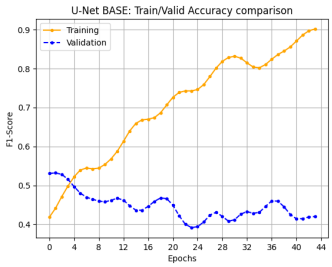
(a) Base model loss



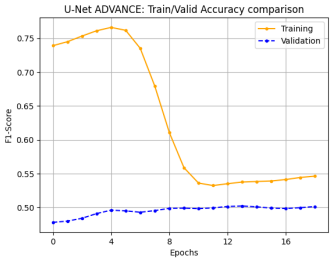
(b) Advance model loss



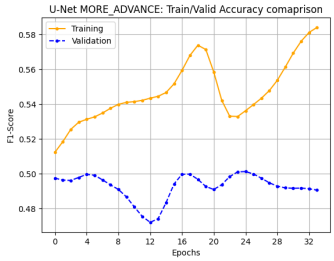
(c) More Advance model loss



(d) Base model f1



(e) Advance model f1



(f) More Advance model f1

Figure 13: The learning curves of the three models: in Fig. (a),(d) it can be seen that the training and validation curves diverge more and more with increasing epochs. The advance model in Fig. (b),(e) limits this problem by increasing the level of feature detail with the introduction of sex and gender embeddings. In Fig. (c),(f) the curves that benefit from the attention mechanism introduced in the model at the level of skip connections.